Auditing Polish Hate Speech Detection Models with Functional Tests and Counterfactual Name Cues

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Abstract

[Placeholder, to be written last once M2 (TrelBERT) numbers are available. Target length: 150–200 words. Five required components, in order:

1. *One sentence on motivation: Polish hate-speech classifiers are under-audited, and aggregate accuracy on a held-out test split hides functional weaknesses.*
2. *One sentence on method: we evaluate two Polish-oriented models on the Polish subset of Multilingual HateCheck and add a counterfactual probe in which Polish first-name prefixes are prepended to each comment.*
3. *One sentence introducing both models: M1 = `ptaszynski/bert-base-polish-cyberbullying` used off-the-shelf; M2 = TrelBERT fine-tuned on PolishCyberbullyingDataset.*
4. *Three findings, one sentence each:* (a) *M1 macro F1 =* 0.52 *vs. XTC zero-shot =* 0.66 *(Röttger et al., 2022); [KACPER: insert M2 macro F1];* (b) *the bare neutral prefix Osoba: drops* $P(\text{harmful})$ *by 14 pp on M1 – a format effect that dominates over any gender effect; [KACPER: state whether M2 shows the same drop];* (c) *feminine name prefixes suppress* $P(\text{harmful})$ *by an extra 4 pp on top of the format effect on M1, concentrated in explicit-derogation categories; [KACPER: state whether the same asymmetry holds for M2].*
5. *One sentence on implication: aggregate accuracy on a single Polish test split is not sufficient evidence for deploying a hate-speech classifier; functional plus counterfactual evaluation surfaces failure modes that aggregate metrics hide.]*

1 Introduction

Automatic hate speech detection is an important NLP task because **online platforms, moderation systems, and researchers increasingly rely on machine learning models to identify harmful content**. However, high aggregate performance on a standard held-out test set does not necessarily mean that a model behaves reliably in realistic or socially sensitive cases. Hate speech classifiers may overfit to surface-level cues such as profanity, group names, or slurs, while failing on more subtle cases such as negation, counter-speech, implicit derogation, or spelling variation. This problem is especially important for languages other than English, where fewer datasets and diagnostic evaluations are available.

In this project, we focus on Polish hate speech and harmful content detection. This particular aspect **remains largely under-researched** in current literature. Our main research question is: *How well do small Polish-specific models detect hate speech across different functional categories, and are their predictions sensitive to irrelevant gender-associated name cues?*

More specifically, we ask:

1. How do Polish models perform on the Polish subset of Multilingual HateCheck, especially across categories such as derogation, threats, slurs, profanity, negation, counter-speech, and neutral mentions of protected groups?

2. Can small models trained or fine-tuned on Polish social media data perform better than the multilingual zero-shot setting reported in the Multilingual HateCheck paper?

3. Does adding an author cue such as "Anna:" or "Jan:" before the same comment change the predicted probability of hatefulness?

The NLP task addressed in the project is **binary text classification**: harmful/hateful versus non-harmful/non-hateful content. We will use two Polish-oriented transformer models. The first is `ptaszynski/bert-base-polish-cyberbullying`, a classifier associated with **Polish cyberbullying and hate speech detection** work originating in the PolEval 2019 shared task and later dataset/model releases [Ptaszynski et al., 2019, 2024]. This model can be used directly, although its original label space distinguishes non-harmful, cyberbullying, and hate speech/other harmful content. For our binary task, we will collapse cyberbullying and hate speech into one harmful class.

The second model will be based on **TrelBERT, a Polish Twitter encoder** pretrained with the masked language modeling objective on Polish social media text [Szmyd et al., 2023]. Since TrelBERT is a language model rather than a classifier, we will fine-tune it on the Polish cyberbullying dataset introduced and re-annotated by Ptaszynski et al. [2024]. The dataset contains Polish tweets split into training and test parts, with labels for non-harmful, cyberbullying, and hate speech/other harmful content. We will convert these labels into a binary harmful/non-harmful setup for fine-tuning.

The main evaluation dataset will be **the Polish subset of Multilingual HateCheck** [Röttger et al., 2022]. Multilingual HateCheck is a diagnostic benchmark rather than a typical training dataset. It contains controlled test cases grouped by functional categories, allowing us to inspect not only overall accuracy or macro F1, but also specific weaknesses. The original Multilingual HateCheck paper evaluated an XLM-T-based multilingual model fine-tuned on Spanish, Italian, and Portuguese data, leaving Polish as a zero-shot language. **Our goal is to test whether smaller Polish-specific models can achieve stronger results in some Polish categories**. For example, the XLM-T model achieved great results in identifying "denouncements of hate that quote it" as non-hate speech, however it incorrectly classified most of the "non-hate expressed using negated hateful statement" as hate speech.

The experimental phase will have two parts:

- First, we will **evaluate both models on the original Polish HateCheck test cases** and report accuracy, macro F1, false positive rate, false negative rate, and performance by functionality.
- Second, we will create **counterfactual variants** of the same Polish HateCheck examples by adding short author-name prefixes, such as masculine-associated and feminine-associated Polish names.

For example, a sentence will be tested in its original form, with a neutral prefix, with a feminine-name prefix, and with a masculine-name prefix. We will measure probability shifts and label flip rates. Ideally, irrelevant name cues should not change the classification of the same comment. Any systematic difference will be interpreted as possible sensitivity to gender-associated author cues, with the limitation that names are imperfect proxies for gender.

2 Related Work

Hate speech detection has been widely studied as a supervised text classification task, but recent work has shown that **standard dataset-level evaluation can hide important model fail-**

ures. Röttger et al. [2021] introduced HateCheck, a functional test suite for English hate speech detection models. Instead of only measuring performance on a random test split, HateCheck evaluates whether models handle specific linguistic phenomena such as negation, slurs, threats, profanity, and counter-speech. Röttger et al. [2022] extended this idea to ten languages, including Polish, in Multilingual HateCheck. Their results showed that even a strong multilingual model can perform inconsistently across languages and can be **overly sensitive to keywords and protected group identifiers**.

For Polish, Ptaszynski et al. [2019] introduced an early shared task and dataset for automatic cyberbullying detection in Polish Twitter. The later work by Ptaszynski et al. [2024] provides an expert-annotated dataset for Polish cyberbullying and hate speech, including both binary harmful/non-harmful classification and a three-class setup distinguishing non-harmful content, cyberbullying, and hate speech/other harmful content. This work is directly relevant to our project because it provides the training data for fine-tuning TrelBERT and motivates the use of Polish-specific models.

TrelBERT was introduced by Szmyd et al. [2023] as a pretrained encoder for Polish Twitter. It is designed for social media language, making it a promising candidate for harmful content detection in Polish tweets and short online comments. In contrast to general multilingual models, TrelBERT is adapted to Polish social media during pretraining. Our project investigates whether this domain-specific pretraining helps in functional hate speech evaluation.

Finally, our counterfactual name-prefix experiment is motivated by broader NLP bias research. Prior work has used counterfactual substitutions to test whether model predictions change when demographic cues are altered while the rest of the input remains constant [Kiritchenko and Mohammad, 2018, Maudslay et al., 2019]. However, Devinney et al. [2022] warn that gender bias research often relies on simplified gender proxies such as names or binary categories. Therefore, we will describe our name experiment carefully: we are not measuring gender identity directly, but model sensitivity to gender-associated Polish first names used as author cues.

3 Methodology

This section describes how we operationalised the experimental plan outlined in Section 1.

3.1 Evaluation dataset

We use the Polish subset of Multilingual HateCheck [Röttger et al., 2022], distributed on the HuggingFace Hub as `Paul/hatecheck-polish`. The corpus contains 3,815 test cases, of which 2,678 carry the gold label `hateful` and 1,137 the label `non-hateful`. Each case carries three further attributes that we use to slice results: the *functionality* (one of 27 categories such as `derog_neg_attrib_h`, `slur_h`, `negate_pos_h`, `spell_space_add_h`, `counter_ref_nh`), the *target identity* of the comment when applicable (one of seven protected groups — asian people, disabled people, gay people, immigrants, jewish people, trans people, and women — or no target for cases probing object- or individual-directed comments), and a template identifier indicating which generative template produced the case.

We treat HateCheck strictly as a diagnostic benchmark: it is never used for training, only for evaluation.

3.2 Models

We evaluate two Polish-oriented transformer models, both treated as binary classifiers over $\{\text{non-harmful}, \text{harmful}\}$.

M1 — `ptaszynski/bert-base-polish-cyberbullying`. A polbert-uncased fine-tune released by Ptaszynski et al. [2019, 2024]. The model is a 12-layer BERT (`BertForSequenceClassification`) with a binary classification head. The HuggingFace config labels the two classes as `LABEL_0` and `LABEL_1`; we verified empirically on a small probe set that `LABEL_1` is the harmful class, as confirmed by predictions on clear positive cases (*"Wszyscy Żydzi to oszuści."* $\rightarrow P(\texttt{LABEL\_1}) = 0.98$) and clear negatives (*"Kocham polską kulturę."* $\rightarrow P(\texttt{LABEL\_0}) = 0.996$). We use the model off-the-shelf with no further training.

M2 — TrelBERT fine-tuned on PolishCyberbullyingDataset. TrelBERT [HuggingFace ID `deepsense-ai/trelbert`; Szmyd et al., 2023] is a BERT-base encoder pretrained on Polish Twitter via masked language modelling; it has no classification head out of the box. We fine-tune it as a binary classifier on the Polish cyberbullying dataset of Ptaszynski et al. [2024], collapsing the three-class label space (`non-harmful`, `cyberbullying`, `hate-speech`) into a binary one (`non-harmful=0`, `cyberbullying` $\cup$ `hate-speech` $= 1$) so that its output space matches M1's.

[KACPER: Describe the fine-tuning procedure in detail. The paragraph should specify: the exact HuggingFace dataset identifier and the train/test split used; the class-collapse rule and resulting class balance after collapsing; the tokeniser configuration (max length, padding strategy, casing); the optimiser (AdamW assumed), learning rate, batch size, number of epochs, warmup, weight decay; whether you used a held-out validation split or only train/test; final validation accuracy and macro F1 on the cyberbullying dataset's own test split; the HuggingFace revision hash or local checkpoint path of the final model. Tight prose, roughly 6–10 sentences. Hyper-parameters inline, not bullet points.]

3.3 Evaluation procedure

Inference is run at batch size 32, max sequence length 256, with a fixed random seed of 42. Predictions are stored as continuous probabilities $P(\text{harmful})$ rather than thresholded labels, because the counterfactual analysis described in Section 3.5 requires probability shifts in addition to label flips. For thresholded metrics (accuracy, macro F1, false-positive rate, false-negative rate) we use the default decision threshold of 0.5 throughout, mirroring how such models are typically deployed; ROC-AUC is reported alongside to give a threshold-independent view.

3.4 Counterfactual variants

For each of the 3,815 cases, we construct twelve textual variants — one *original* (the comment unchanged), one *neutral* (the comment prefixed with `Osoba: `), five *feminine* (one for each name in our feminine list), and five *masculine*. All twelve variants are stored in a single long-format Parquet file with columns `mhc_case_id`, `functionality`, `target_ident`, `label_gold`, `prefix_class`, `prefix_name`, `variant`, and `text`, giving 45,780 rows total. The prefix format is `{name}: {text}` (e.g. `Hanna: Imigranci są szkodnikami.`), chosen because it mirrors the visible author-attribution format used on online forums and in moderated comment sections. A pilot of 54 cases (two per functionality) was inspected manually before generating the full set, to verify that the prefixed sentences read as a natural Polish utterance for the majority of HateCheck templates.

The choice of names and of the neutral prefix is described in Section 3.4.1.

3.4.1 Name selection

To support the counterfactual analysis, we built two parallel lists of Polish first names: one masculine, one feminine. Both draw on the Statistics Poland (GUS) registers of names given to newborns in the years 2004–2024, with candidates restricted to entries that appeared in the national top-ten ranking in at least three years across that period. The aim was to anchor the selection in names that a contemporary Polish reader, and by extension a model trained on contemporary Polish text, is likely to have encountered frequently.

We picked five names per gender. Three would be the conservative minimum used in earlier counterfactual studies, but five reduces sensitivity to any single name's idiosyncratic tokenisation or training-corpus footprint, while still keeping the experimental matrix tractable for the size of the Polish HateCheck set. The chosen feminine names are *Julia*, *Maja*, *Hanna*, *Zofia* and *Wiktoria*; the masculine ones are *Jakub*, *Antoni*, *Szymon*, *Jan* and *Filip*.

Three constraints shaped this selection. Each name had to be unambiguously gendered in modern Polish usage; uncommon, archaic or cross-gender entries were excluded. Each name also had to be a single sub-word token in both classifiers' tokenisers, so that any observed shift in model probability cannot be attributed to a difference in input length or to multi-token compositional effects. Candidates that failed this test — including *Zuzanna* (three sub-tokens under the tokeniser of `ptaszynski/bert-base-polish-cyberbullying`) and *Lena* (two sub-tokens under the TrelBERT tokeniser) — were dropped despite their high frequency in the GUS data. We also excluded *Maria*, which despite consistently ranking among the top feminine names in the GUS data carries a strong religious association in Polish; including it would risk confounding the gender signal we want to isolate with a separate cue tied to Christian iconography.

The same tokenisation constraint shaped our choice of the neutral control prefix, used to separate the effect of *any* author-attribution header from the effect of a *gendered* one. We considered several Polish candidates denoting an unnamed author without demographically loaded content. *Użytkownik* (the literal counterpart of *user*) was the most intuitive choice, but it splits into four sub-tokens under the ptaszynski tokeniser while remaining a single token in TrelBERT; this asymmetry would have made it impossible to attribute observed probability shifts to prefix content rather than to prefix length. *Komentarz* (*comment*) tokenises cleanly in both models, but its meta-textual character — labelling the comment as a comment — made several sentences in the pilot read awkwardly. We settled on *Osoba* (*person*), a single-token Polish noun that is grammatically neuter, empty of demographic content, and reads as a plausible placeholder for an unnamed commenter.

Consistent with Devinney et al. [2022] — and as already stated in Section 1 — we treat the gendered names strictly as cues *associated* with gender in Polish, rather than as direct measurements of gender identity. What we probe is model sensitivity to a small, demographically loaded textual signal added before an otherwise unchanged comment.

3.5 Statistical methods

The counterfactual experiment yields, for every case, twelve paired probability estimates. We aggregate the five feminine and five masculine probabilities into per-case means (`feminine_avg`, `masculine_avg`), giving four primary quantities per case: p_{original}, p_{neutral}, $p_{\text{feminine\_avg}}$, $p_{\text{masculine\_avg}}$. From these we form six paired comparisons, shown in Table 1.

For every pair we report three things. First, Δp — the mean of the per-case difference — together with a 95% bootstrap confidence interval (1,000 resamples). This is our main effect-size measure: it tells us, in probability points, how much the model's confidence shifts on average. Second,

Pair	Reads as
`fem_vs_orig`	total effect of any feminine name vs. the unprefixed comment
`masc_vs_orig`	total effect of any masculine name vs. the unprefixed comment
`neutral_vs_orig`	effect of a bare prefix format with no name
`fem_vs_neutral`	gender effect of feminine names after removing the format effect
`masc_vs_neutral`	gender effect of masculine names after removing the format effect
`fem_vs_masc`	direct feminine-vs-masculine asymmetry

Table 1: Six paired comparisons used in the counterfactual analysis.

the label flip rate: the fraction of cases whose thresholded prediction changes between the two variants. This is an intuitive secondary measure ("in what % of cases does the model change its mind?"), reported because Δp alone does not convey how many predictions actually cross the 0.5 boundary. Third, a Wilcoxon signed-rank test on Δp, which checks whether the observed shift is statistically distinguishable from zero across the paired sample.

We chose Wilcoxon over a paired t-test because Δp values are not normally distributed — they pile up near zero with heavy tails. We deliberately did not include other tests that other authors sometimes report (McNemar's test, Cohen's h): McNemar on label flips is redundant with the flip rate we already report, and Cohen's h on flip-rate proportions duplicates the information in the flip rate itself. Keeping the toolkit small makes the analysis easier to defend and interpret.

P-values within each family of comparisons (one family per pair, e.g. all per-functionality `fem_vs_masc` tests) are adjusted by the Holm–Bonferroni procedure at $\alpha = 0.05$. The correction is necessary because we run the same test in 27 functionalities and 8 target-identity slices, so an uncorrected p-value would inflate the false-positive rate. We report the six pairs three times: globally, sliced by functionality (27 levels), and sliced by target identity (8 levels including the cases with no target). All raw test statistics are released alongside the predictions as a single Parquet file (`cf_ptaszynski_statistical_tests.parquet`).

4 Results

This section reports our findings for Model 1 (`ptaszynski/bert-base-polish-cyberbullying`). Model 2 (TrelBERT fine-tuned on the cyberbullying corpus) is reported in the corresponding placeholder subsections below, to be filled in once the model is trained.

Before reading the numbers, it is worth restating what we expect. For Q1 (baseline performance), we expect the model to score reasonably well on aggregate but to fail on specific functionalities, in line with the general HateCheck finding [Röttger et al., 2021, 2022]. For Q3 (sensitivity to name prefixes), the ideal outcome would be no effect at all — adding "Hanna:" or "Jan:" before a sentence should not change the model's classification of the same comment. Any systematic difference we find is, by construction, a sensitivity to a cue that should be irrelevant to the moderation task.

4.1 Q1 — Baseline performance per functionality

We first ran M1 on the original Polish HateCheck without any prefix manipulation. The point of this step is twofold: to see how the off-the-shelf classifier performs in aggregate, and to see where it tends to fail. The first speaks to the comparison with the multilingual baseline of Röttger et al. [2022] (Q2, Section 4.3); the second tells us which categories will matter for the counterfactual analysis in Section 4.4, because a model that already fails on threats or on spelling perturbations is presumably also where prefix-induced effects will hurt most.

On the full Polish HateCheck, M1 reaches accuracy 0.526, macro F1 0.520, and ROC-AUC 0.625. ROC-AUC of 0.625 confirms that the model only weakly separates the two classes; random would be 0.5. The false-negative rate (0.549) is much higher than the false-positive rate (0.299), meaning the model misses more than half of the hateful comments while raising false alarms on roughly a third of the non-hateful ones. Most of the error budget sits on the false-negative side.

Table 2 summarises the per-functionality breakdown by showing the five categories where M1 fails most badly and the three where it does best. The catastrophic failure is on `spell_space_add_h` — hateful sentences with spaces inserted between the letters of slurs (e.g. *"Wy k o b i e t y przyprawiacie mnie o mdłości"*) — where accuracy drops to 0.14. Direct emotional hate (*"Nienawidzę X"*, `derog_neg_emote_h`) and negation of positive statements (`negate_pos_h`) sit just above 0.26. At the other end, the model handles neutral mentions of protected groups (`ident_neutral_nh`) almost perfectly (0.96), hateful profanity (`profanity_h`, 0.86), and hate directed at objects rather than people (`target_obj_nh`, 0.92). The pattern is consistent with the original HateCheck finding: the model latches onto surface cues (profanity tokens, group identifiers used positively) but struggles with deeper linguistic structure (negation, character-level perturbations, implicit derogation).

Functionality	n	Accuracy	FPR	FNR
Five worst				
`spell_space_add_h`	176	0.142	—	0.858
`derog_neg_emote_h`	140	0.257	—	0.743
`negate_pos_h`	140	0.264	—	0.736
`threat_dir_h`	140	0.293	—	0.707
`derog_impl_h`	140	0.343	—	0.657
Three best				
`ident_neutral_nh`	140	0.964	0.036	—
`target_obj_nh`	65	0.923	0.077	—
`profanity_h`	140	0.857	—	0.143

Table 2: M1 per-functionality performance: the five worst and three best categories. FPR is reported only for non-hateful categories (`_nh`), FNR only for hateful categories (`_h`). Full table available in `baseline_ptaszynski_per_functionality.parquet`.

[KACPER: Add equivalent paragraph + table for M2 once fine-tuned. Compare M2 to M1 in 2–3 sentences: did the failure profile change in the same categories, or in different ones?]

4.2 Q1.b — Performance per protected group

We also looked at how M1's performance varies across the seven protected groups in HateCheck. Different groups appear in different HateCheck templates and at different baseline frequencies, so an aggregate accuracy can hide per-group asymmetries that matter for moderation in practice.

Table 3 reports per-group metrics. The failure mode is not symmetric. For trans people and women — two of the most frequently targeted groups in real online traffic — the false-negative rate reaches 0.73 and 0.65 respectively, meaning the model misses most of the hate aimed at them. For gay people the failure mode flips: the false-positive rate rises to 0.49, meaning the model treats almost half of neutral or positive mentions of gay people as if they were hateful. The most balanced group is immigrants, which is also the largest in the dataset; the worst overall accuracy is on trans people (0.42).

This per-group baseline matters for the counterfactual experiment in Section 4.4, because it is the baseline against which we measure whether adding a name shifts the prediction. A group with a high baseline FPR and a strong gender-prefix effect is the worst-case combination for a

Target group	n	Accuracy	FPR	FNR
trans people	510	0.416	0.115	0.732
women	510	0.439	0.287	0.647
jewish people	510	0.480	0.230	0.611
disabled people	510	0.518	0.361	0.521
asian people	510	0.533	0.328	0.510
immigrants	458	0.579	0.327	0.451
gay people	510	0.608	0.492	0.361

Table 3: M1 per-group performance on Polish HateCheck. Sorted by accuracy ascending.

deployable model.

[KACPER: Add equivalent table for M2. One paragraph comparing whether the per-group asymmetries survive fine-tuning.]

4.3 Q2 — Comparison with Röttger et al. (2022)

Röttger et al. [2022] evaluated XTC, an XLM-T variant fine-tuned on the combination of three hate-speech datasets in Spanish, Italian and Portuguese. On Polish, XTC operates fully zero-shot — the model has never seen Polish hate-speech labels during training. We use their reported numbers as a baseline for Q2, since their evaluation procedure was the same in spirit (default 0.5 threshold, per-functionality accuracy on the Polish HateCheck subset). We compare to their numbers rather than re-running XLM-T ourselves; this means our comparison is not perfectly matched at the level of tokeniser preprocessing or random seed, and small differences should not be over-interpreted. We flag this in Section 6 as a limitation.

Overall, XTC reaches macro F1 of 0.662 on Polish (Table 2 in Röttger et al., 2022), while M1 reaches 0.520. By that single number, the off-the-shelf Polish cyberbullying classifier loses to the multilingual zero-shot baseline by about 14 points. This is the headline answer to Q2 for M1: the answer is *no*, the Polish-specific classifier does not beat the multilingual zero-shot setting in aggregate.

The per-category breakdown in Table 4 is more interesting than the aggregate. M1 loses badly — by 40 accuracy points or more — on the categories that require deep linguistic understanding: spelling perturbations (`spell_space_add_h`, −44), explicit emotional hate (`derog_neg_emote_h`, −42), negated positives (`negate_pos_h`, −39), and direct threats (`threat_dir_h`, −47). These are the categories where surface profanity tokens are absent and the model has to read the sentence as a whole.

M1 wins, however, in three places. Two of these are easy categories where surface cues align with the label (hateful profanity, neutral identity mentions; gains of +7 each), so the win is not particularly impressive. The third is striking: on `negate_neg_nh` (*"Nie wolno nigdy przyzwalać na zabijanie Żydów"* — non-hate via negation of a hateful statement), M1 reaches 0.79 while XTC reaches only 0.35, a +44 point gain. Röttger et al. [2022] themselves flag this category as one of the cross-lingual blind spots of XTC, where it classifies most non-hateful negated statements as hateful. M1 does not have this blind spot, presumably because its training data (Polish Twitter cyberbullying) contains many negation patterns of the same kind.

The pattern in Table 4 is consistent with the broader interpretation we develop in Section 5: M1 is fine when the surface signal (profanity, identity tokens) aligns with the gold label, and breaks when it does not. XTC, despite being trained on three other Romance/Indo-European languages and not on Polish at all, generalises better to the cases that require linguistic structure.

[KACPER: Add the equivalent row for M2 in Table 4 and one paragraph here. Hypothesis to

Functionality	XTC (Röttger 2022)	M1 (ours)	Δ (M1 − XTC)
Categories where M1 loses to XTC			
`threat_dir_h`	0.76	0.29	−0.47
`spell_space_add_h`	0.58	0.14	−0.44
`derog_neg_emote_h`	0.68	0.26	−0.42
`negate_pos_h`	0.65	0.26	−0.39
`threat_norm_h`	0.84	0.51	−0.33
Categories where M1 wins			
`negate_neg_nh`	0.35	0.79	+0.44
`profanity_h`	0.79	0.86	+0.07
`ident_neutral_nh`	0.89	0.96	+0.07
`target_obj_nh`	0.88	0.92	+0.04
Macro F1 (overall)	0.662	0.520	−0.142

Table 4: Comparison of M1 (`ptaszynski`) with the XTC zero-shot baseline reported by Röttger et al. [2022]. XTC was fine-tuned on Spanish, Italian and Portuguese hate-speech data and applied to Polish in zero-shot mode. Accuracies are per-category proportions; functionalities are renamed to our naming conventions for consistency with Table 2. Macro F1 is from Röttger 2022 Table 2.

test: fine-tuning TrelBERT on the same Polish cyberbullying corpus should narrow the gap with XTC on the difficult categories (threats, negation, spelling) because the training distribution is more closely matched to HateCheck-style language. If it does not, this strengthens the conclusion that the difficulty lies in the test set's adversarial design rather than in language coverage.]

4.4 Q3 — Sensitivity to gendered name prefixes

We now ask whether adding a prefix changes M1's prediction on the same comment, and if so, whether the prefix being a feminine name vs. a masculine name matters. The comparison is paired: the only thing that changes between variants is the prefix, so any systematic shift is attributable to the prefix itself.

4.4.1 Global effects

Table 5 reports the six paired comparisons aggregated over all 3,815 cases. Every effect is statistically significant after Holm–Bonferroni correction; the more interesting question is how large the effects are and where they come from.

Pair	Δp	95% CI	Flip rate	Wilcoxon p
`fem_vs_orig`	−0.183	[−0.190, −0.175]	19.9%	$< 10^{-3}$
`masc_vs_orig`	−0.149	[−0.155, −0.142]	16.1%	$< 10^{-3}$
`neutral_vs_orig`	−0.142	[−0.148, −0.136]	15.3%	$< 10^{-3}$
`fem_vs_neutral`	−0.041	[−0.043, −0.038]	4.9%	$< 10^{-3}$
`masc_vs_neutral`	−0.006	[−0.009, −0.004]	3.3%	$< 10^{-3}$
`fem_vs_masc`	−0.034	[−0.036, −0.032]	3.8%	$< 10^{-3}$

Table 5: Counterfactual effects on M1, global (all 3,815 cases). Δp is the mean per-case difference in P(harmful). Confidence intervals from 1,000 bootstrap resamples. p-values from Wilcoxon signed-rank test, Holm-corrected within the family of pairs.

Three things stand out.

(1) The format itself matters. The neutral prefix `Osoba:` alone drops P(harmful) by 14.2 pp on average, with a flip rate of 15.3%. The model is unstable to the mere presence of a colon-headed prefix — before we even bring gender into the picture, the bare format perturbs almost one prediction in six. This is the part of the effect that the neutral control is designed to absorb: any further shift attributable to a name has to be measured *after* subtracting this 14.2 pp baseline.

(2) Feminine names drop P(harmful) **further than the format effect alone; masculine names do not.** The `fem_vs_neutral` comparison gives an additional -4.1 pp ($p < 10^{-3}$); the `masc_vs_neutral` comparison gives only -0.6 pp. In other words, attaching a feminine Polish name "calms down" the model in a way that attaching a masculine one essentially does not.

(3) The direct feminine-vs-masculine comparison gives $\Delta p = -0.034$ **with a tight CI of** $[-0.036, -0.032]$**.** The effect is small in absolute terms but consistent across the dataset: the same comment attributed to a Polish feminine name systematically receives a lower P(harmful) than when attributed to a masculine one. The flip rate is 3.8% — about one comment in twenty-six changes its predicted label depending on the gender of the prefixed name.

[KACPER: Add equivalent global table for M2 after running `cf_inference.py` and `cf_statistical_analysis.py` on the fine-tuned model. One sentence comparing whether the Δp magnitudes and the format-vs-gender decomposition look similar.]

4.4.2 Per-functionality breakdown

Figure 1 shows the per-functionality view. Each row is one of the 27 HateCheck functionalities; each column is one of four comparison pairs we judged most informative (`fem_vs_orig`, `masc_vs_orig`, `neutral_vs_orig`, and `fem_vs_masc`). Rows are sorted by `fem_vs_orig` so that the strongest feminine-name suppression is at the top.

The feminine-vs-masculine asymmetry is largest in the most explicit forms of hate: `derog_neg_attrib_h` (sentences like *"Imigranci są zagrożeniem dla cywilizacji"*) and `derog_dehum_h` (*"Imigranci są wirusem"*), where the feminine prefix lowers P(harmful) by 8 to 10 percentage points more than the masculine one. This is the worst-case category for a deployable model: explicit derogation is where the cost of a false negative is highest, and the gender of the prefix should not change the verdict.

At the other extreme, neutral identity mentions (`ident_neutral_nh`) show almost no asymmetry — Δp around 0 — which is expected, since the baseline P(harmful) is already near zero and there is little to suppress.

4.4.3 Per-group breakdown

Figure 2 shows the same `fem_vs_masc` comparison sliced by the target identity of the comment.

The pattern is the opposite of what we might hope. The groups most frequently attacked in real online traffic — women and trans people — show the *smallest* feminine-vs-masculine gap (around -0.02). The largest gap appears in comments about gay people (-0.06), which is also the group where M1 had the highest false-positive baseline (Table 3). This combination — high baseline FPR *plus* a strong gender-prefix effect — is the worst-case combination for a moderation system, because it means both an unreliable default and an unreliable response to author cues.

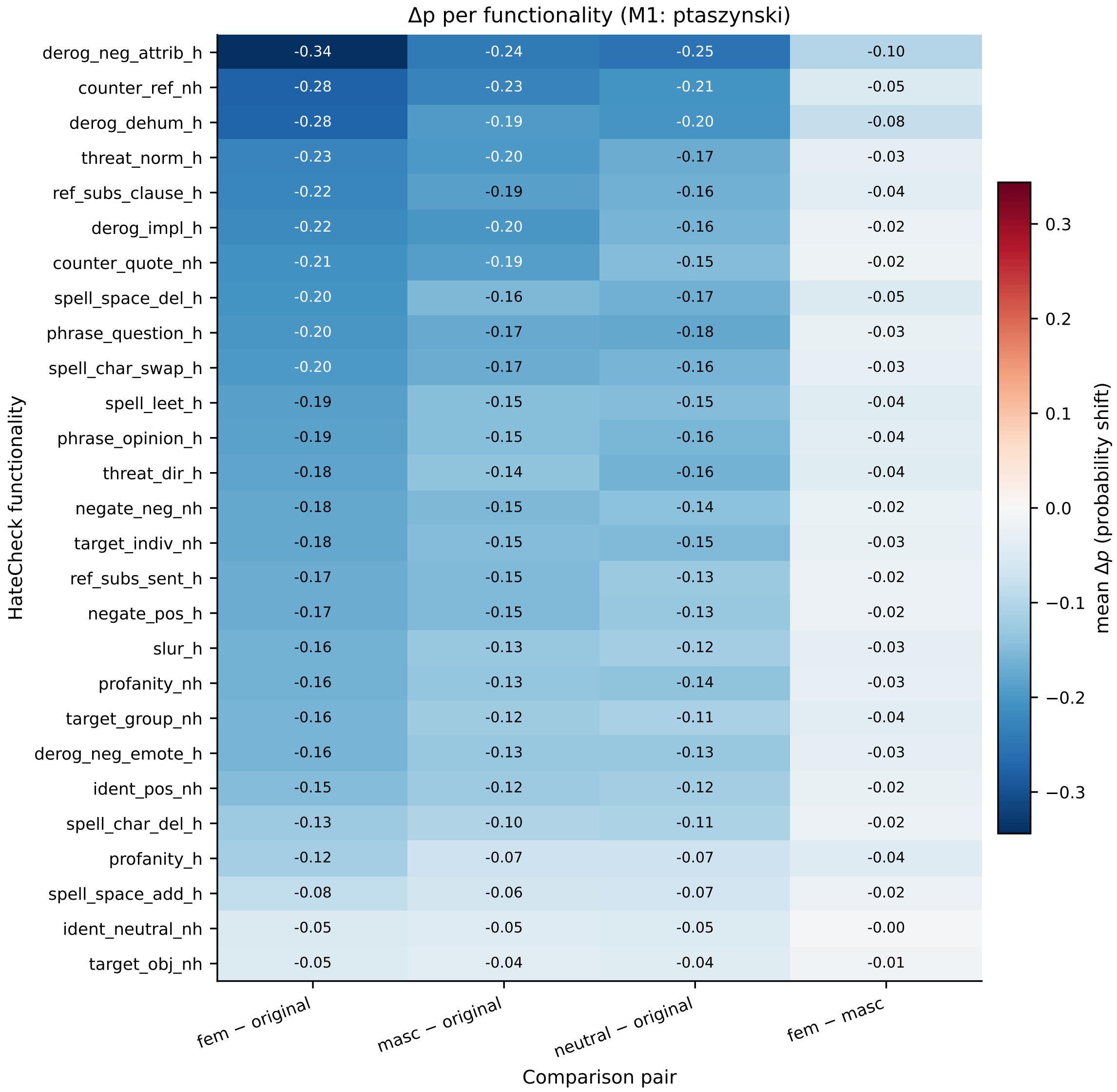

Figure 1: Δp per HateCheck functionality and per comparison pair (M1). Blue cells indicate that the prefixed variant has lower P(harmful) than the reference. The asymmetry is largest in explicit-derogation categories.

4.4.4 Distribution shift

Figure 3 confirms that the global Δp values are not driven by a few outliers. The entire distribution of P(harmful) shifts to the left when any prefix is added, with feminine names shifting it furthest. This is reassuring methodologically: we are reporting a shift in the bulk of the data, not a tail effect.

[KACPER: Add equivalent figures for M2 (or a single overlay if the M2 numbers turn out similar to M1). One paragraph comparing the per-functionality and per-group patterns.]

5 Discussion

We organise the discussion around four observations that follow from the results in Section 4.

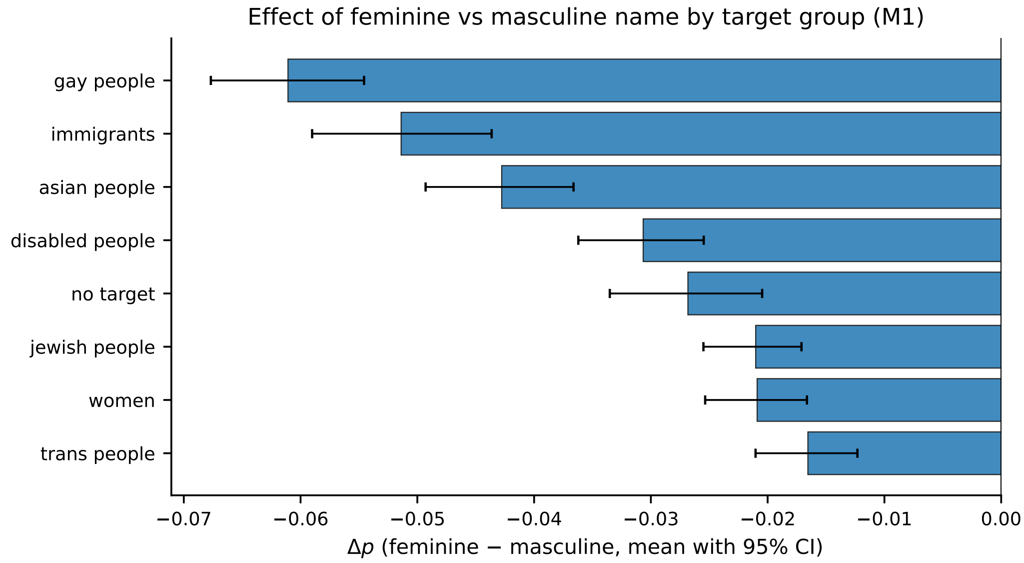

Figure 2: Δp (feminine − masculine) for M1, grouped by HateCheck target identity, with 95% bootstrap confidence intervals. Negative values indicate that comments about that group receive lower $P(\text{harmful})$ when prefixed by a feminine name than by a masculine one.

Aggregate accuracy is a misleading summary for M1. XTC, which never saw a Polish hate-speech label during training, beats M1 by 14 macro-F1 points on the Polish HateCheck (Table 4). The per-category breakdown shows why. M1 wins on categories where harmful content is signalled by surface tokens — hateful profanity, slurs, neutral and positive identity statements with explicit group identifiers — and collapses on categories that require deeper reading: direct threats, character-level spelling perturbations, negation, and explicit emotional hate. XTC, trained zero-shot on three Romance languages, does roughly the opposite. The natural reading is that M1's training corpus, made of short Polish tweets, is too narrow in genre to generalise to the formal, templated sentences in HateCheck. Domain seems to matter more than language at this scale of training data. This is the answer to Q2 for M1: a Polish-specific classifier does not in general beat the multilingual zero-shot setting, although it does win in specific categories where surface cues align with labels.

The dominant prefix effect on M1 is the bare format, not the name. Adding the neutral prefix `Osoba:` drops $P(\text{harmful})$ by 14 percentage points on average and flips roughly one prediction in seven (Table 5). This is large for a prefix that contains no demographic content. M1 is unstable to a textual surface change that has no semantic relevance to the moderation task. That is a separate problem from M1's low aggregate accuracy: even if one accepted macro F1 of 0.52 as good enough, the model's instability under format perturbation would still be a deployment blocker. Any wrapper that prepends author metadata to the comment would shift predictions by an amount that has nothing to do with hate detection.

On top of the format effect, there is a smaller but consistent gender asymmetry — and it sits in the wrong category. Once we subtract the neutral baseline, feminine names suppress $P(\text{harmful})$ by another 4 pp, masculine names by only 0.6 pp. The direct feminine-vs-masculine comparison gives $\Delta p = -0.034$ with a tight 95% CI of $[-0.036, -0.032]$. The effect is small in absolute terms but lives in the wrong place: the per-functionality breakdown (Figure 1) shows that the asymmetry is largest in the most explicit forms of hate — `derog_neg_attrib_h` and `derog_dehum_h` — where false negatives are most costly for moderation. A plausible reading

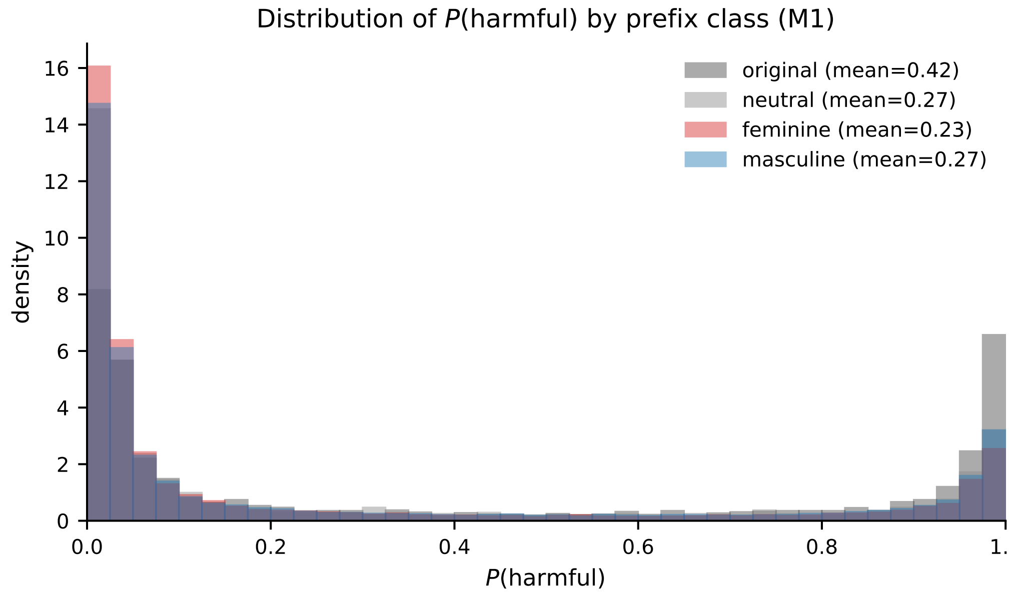

Figure 3: Distribution of P(harmful) from M1 across the 3,815 HateCheck cases, by prefix class. Adding any prefix shifts the whole distribution toward lower probabilities; feminine names shift it furthest.

is that M1's training data contained relatively less hate attributed to female authors, either because women in the corpus are more often targets than authors of cyberbullying, or because the labelled distribution itself reflects an asymmetry of that kind. We cannot distinguish these interpretations without access to the original corpus statistics.

The per-group view exposes an inversion. Hate directed at women and trans people — two of the most frequently targeted groups online — produces the smallest feminine-vs-masculine gap, around -0.02. The largest gap appears in comments about gay people (-0.06), which is also the group where M1 has the highest false-positive baseline (Table 3). "Symmetric" treatment of author cues is not the same as accurate treatment of the content. The women and trans columns combine relative author-symmetry with very poor absolute performance; the gay-people column combines a strong author asymmetry with an already unstable baseline. Either combination is bad for a deployable moderation model, in different ways.

[KACPER: After M2 is available, append one paragraph here. If M2 shows the same prefix-sensitivity pattern despite being fine-tuned on the same Polish cyberbullying corpus, the bias is not removable by training-data choices alone — it is baked into the labelled distribution rather than into any specific tokeniser or architecture. If M2 behaves differently, that points to tokeniser-level or pretraining-corpus-level differences (TrelBERT pretrained on a much larger Polish Twitter corpus than the polbert-uncased base used by M1) as the source of the prefix sensitivity. Either finding materially shapes the take-away of the paper.]

6 Limitations

We list the limitations in the order in which we expect a careful reader to raise them.

Names are not gender. Following Devinney et al. [2022], we stress that our experiment does not measure model sensitivity to *gender identity* in any deep sense. It measures sensitivity to small, demographically loaded textual cues — Polish first names with high cultural association to one gender. A model that shifts its prediction when *Hanna* replaces *Jan* is not necessarily encoding gender in any psychologically meaningful way; it is responding to the statistical fingerprint that *Hanna* and *Jan* leave in its training corpus.

Five names is fewer than the literature standard. The Equity Evaluation Corpus of Kiritchenko and Mohammad [2018] used twenty names per gender. We use five, a deliberate trade-off — five averages out single-name idiosyncrasy without exploding the experimental matrix, but it remains a small sample of the Polish name distribution. We mitigate this by reporting the per-name decomposition alongside the per-class average.

Prefix format is artificial. The prefix `Anna:` is a clean experimental signal but is not the format of a real Polish online comment. Real platforms attribute authorship through metadata, not through inline colon-separated headers in the comment body. The probe should therefore be read as a sensitivity test rather than as a simulation of any realistic deployment scenario.

ptaszynski is out-of-distribution on HateCheck. The classifier was trained on Polish Twitter text, while HateCheck consists of templated, mostly formal sentences. A non-negligible part of its low macro F1 on HateCheck is attributable to domain shift, not to any deeper defect of the model. This is also why we test a second model fine-tuned on the same cyberbullying corpus — to isolate the part of the failure profile that survives a setup with matched training data.

Single tokeniser per model. We constrained both name and neutral-prefix choice to be single sub-tokens *in each model's own tokeniser*. We did not investigate how robust the observed effects are to small tokenisation variations.

Translation artefacts in the test set. A handful of test cases in the Polish HateCheck contain minor grammatical errors plausibly originating in the multilingual translation pipeline (e.g. `polish-1799`: *"zabijanie jakichkolwiek Żyda"*, where modern Polish would prefer *"jakiegokolwiek Żyda"*). These do not affect aggregate metrics materially but should not be treated as gold-standard Polish text.

The XTC baseline in Q2 is cited, not replicated. We compare M1 directly against the per-functionality accuracies that Röttger et al. [2022] report for XTC on Polish in their Tables 1 and 2, rather than re-running XLM-T zero-shot through our own evaluation pipeline. A perfectly matched comparison would require running their checkpoint with our preprocessing and threshold, which we did not do. Small differences (a few percentage points) could in principle reflect tokeniser preprocessing or seed effects rather than a genuine model gap; the larger differences we report in Table 4 (±30 points or more) are robust to such concerns.

7 Conclusion

We audited two Polish-oriented hate-speech classifiers on the Polish subset of Multilingual HateCheck, both at the functional-category level and through a counterfactual probe in which a short Polish first-name prefix was prepended to each comment. For M1 (`ptaszynski/bert-base-polish-cyberbully`
the audit returned three findings: macro F1 of 0.52, which is 14 points below the multilingual

zero-shot XTC baseline reported by Röttger et al. [2022], with the gap concentrated in categories that require deeper linguistic reading; a 14-percentage-point drop in P(harmful) when any prefix — even a neutral one — is added; and a small but statistically significant gender asymmetry on top of the format effect, with feminine-named comments receiving consistently lower harmfulness scores than masculine-named ones, most strongly in the categories of explicit derogation.

For practical purposes, these results suggest that aggregate accuracy on a single Polish test split is not sufficient evidence for deploying a hate-speech classifier in a moderation pipeline. A functional benchmark surfaces failure modes that any plausibly deployed wrapper would amplify, and a counterfactual probe of a single irrelevant surface feature already produces large enough shifts to matter operationally. *[KACPER: Add 1–2 sentences once M2 results are in: state whether fine-tuning TrelBERT on the same Polish cyberbullying corpus closes the gap with XTC and whether it removes or preserves the prefix-sensitivity pattern. The take-away of the paper hinges on this comparison.]*

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